

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 4: Three-Dimensional Wing Airflow & Induced Drag

Syllabus Source: CAE Oxford (pp. 109-144) | Pass Target: >= 90% AviationExam

1. THREE-DIMENSIONAL AIRFLOW & WINGTIP VORTICES

A real finite wing produces 3D airflow due to spanwise pressure differentials between upper and lower wing surfaces:

- **Spanwise Flow:** Air on the high-pressure lower surface flows **OUTWARD** toward wingtips; air on the low-pressure upper surface flows **INWARD** toward fuselage. At the trailing edge and wingtips, these opposing spanwise flows roll up into trailing **WINGTIP VORTICES**.
- **Induced Downwash (w):** Wingtip vortices trail downstream and induce a strong downward vertical velocity component behind the wing called induced downwash. Downwash tilts the local relative airflow vector downwards!
- **Effective Angle of Attack:** Effective AOA = Geometric AOA - Induced Angle (α_i). Because the relative airflow is tilted downward, the total aerodynamic lift vector tilts **REARWARD**. The rearward-tilted component of lift is **INDUCED DRAG** (Vortex Drag)!

2. INDUCED DRAG FORMULA & WING GEOMETRY FACTORS

Induced drag is the inevitable aerodynamic price of generating lift with a finite-span wing:

- **Induced Drag Coefficient:** $C_{Di} = (C_L^2) / (\pi \times AR \times e)$, where AR = Aspect Ratio, e = Oswald wing efficiency factor (~0.8 to 0.9).
- **Variation with Speed:** Because C_L is inversely proportional to V^2 in level flight ($C_L = W / (q S)$), Induced Drag is **INVERSELY PROPORTIONAL TO SPEED SQUARED**: D_i proportional to $1 / V^2$! Induced drag is massive at low speeds (take-off, climb, landing approach) and becomes negligible at high cruising speeds.
- **Variation with Mass:** Induced drag is directly proportional to the **SQUARE OF AIRCRAFT GROSS MASS**: D_i proportional to W^2 ! A 20% increase in mass increases induced drag by $(1.20)^2 = 44\%$!

Design Factor	Aerodynamic Influence on Induced Drag	Aeronautical Examples & Trade-offs
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Aspect Ratio (AR = b^2 / S)	HIGH Aspect Ratio dramatically REDUCES induced drag (spreads lift over longer wingspan, reducing vortex strength).	Gliders (AR ~ 25 to 30), high-altitude surveillance (U-2). Limited by wing root bending moments and structural weight.
Wing Planform Taper	Elliptical lift distribution produces minimum induced drag ($e = 1.0$). Tapered wings approximate elliptical distribution.	Supermarine Spitfire (pure elliptical); modern airliners use trapezoidal tapered wings with aerodynamic washout.
Winglets / Sharklets	Vertical wingtip fences diffuse tip vortices, converting vortex energy into forward aerodynamic thrust.	Airbus sharklets / Boeing blended winglets reduce fuel burn by 3% to 5% in cruise.
Wing Washout (Twist)	Geometric wing twist (lower incidence at tip than root).	Ensures wing root stalls FIRST , preserving aileron roll control at high angles of attack.

[!] EXAM TRAP (AVIATIONEXAM) - Wake Turbulence & Heavy Jet Vortex Hazard

- When is Wake Turbulence Greatest?: When the generating aircraft is HEAVY, CLEAN (flaps/gear up), and SLOW! - Rotation / Touchdown Points: Vortices begin at nose-wheel rotation and end when nose-wheel touches down. - Avoidance Rule: Rotate BEFORE the preceding heavy aircraft's rotation point, and land BEYOND its touchdown point!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: How does Induced Drag vary with aircraft airspeed in straight and level flight?**

[A] Directly proportional to airspeed [B] Directly proportional to airspeed squared (V^2) [C] Inversely proportional to airspeed SQUARED ($1 / V^2$) [D] It remains constant

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. In level flight, as speed increases, C_L decreases by $1/V^2$. Because induced drag is proportional to C_L^2 , Induced Drag decreases with $1/V^2$, making it dominant at low speeds and tiny at high speeds.

Q2: What is the primary aerodynamic benefit of fitting winglets to modern jet transport aircraft?

[A] To increase maximum take-off mass [B] To reduce Induced Drag by impeding spanwise airflow around wingtips and dissipating wingtip vortices [C] To prevent deep stall [D] To improve rudder authority

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Winglets act as vertical aerodynamic barriers that disrupt the high-to-low pressure spanwise bleed around wingtips, reducing vortex downwash and cutting induced drag by 3-5%.

Q3: Under what flight conditions does a transport aircraft generate the strongest, most hazardous wake turbulence?

[A] Fast, light, flaps fully extended [B] HEAVY, CLEAN (flaps retracted), and SLOW [C] Cruising at high Mach number [D] In a steep dive

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Wake vortex strength is proportional to aircraft weight and inversely proportional to airspeed and wingspan. A heavy aircraft flying slowly in clean configuration generates maximum vortex circulation.